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Example : Ising models

$$\mathcal{H} = -J \sum_{ij} s_i s_j$$

$$s_i = \{-1, +1\}$$

$$-\sum_{ij} J_{ij} s_i s_j$$

$$V(\vec{r}_i, \vec{r}_j) \sim \frac{\sigma}{|\vec{r}_i - \vec{r}_j|}$$

Generative methods

$$X = \{ \underbrace{x_0, x_1 \ldots x_{n-1}}_{\text{i.i.d}} \}$$

$$P(x) = \prod_{i=0}^{n-1} P_i(x_i)$$

parameters

$$p(x_i, h_j; \theta)$$

hidden variable (i.i.d)

Energy-model (Boltzman)

$$P(x_i, h_j; \theta) = \frac{f(x_i, h_j; \theta)}{Z(\theta)}$$

normalization factor

$$\sum_{x_i, h_j} P(x_i, h_j; \theta) = 1$$

$$Z(\theta) = \sum_{x_i \in X} \sum_{h_j \in H} f(x_i, h_j; \theta)$$

Ising model

$$E = -J \sum_{i,j} s_i s_j$$

$$P_i(T) = \frac{e^{-E_i/k_B T}}{Z(T)}$$

$$Z(T) = \sum_j e^{-E_j/k_B T}$$

$$f(x_i, h_j; \theta) = e^{-E(x_i, h_j; \theta)}$$

$$p(x_i, h_j; \theta) = \frac{e^{-E(x_i, h_j; \theta)}}{Z(\theta)}$$

$$\boxed{p(x_i; \theta)} = \sum_{h_j \in H} p(x_i, h_j, \theta)$$